

Big Mountain Ski Resort

Increasing the
revenue



Problem Statement pt. 1

Big Mountain recently has installed a new chairlift which will bring up operating costs by about \$1.5M.

- What predictive model can be made to help decide the course of action for Big Mountain in order to increase revenue to offset the additional operation costs.
- Ski industry data will be used to compare resort features to others in the market.

Problem Statement pt. 2

Insight stakeholders:

- Jimmy Blackburn (Dir. of Ops)
- Alesha Eisen (Database Manager)
- Key data source: We will be pulling from the `ski_resort_data.csv` and look at the adult weekday and weekend values to help make a prediction for Big Mountain's prices by comparing resort features with others.

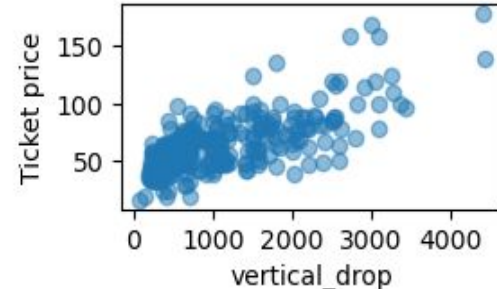
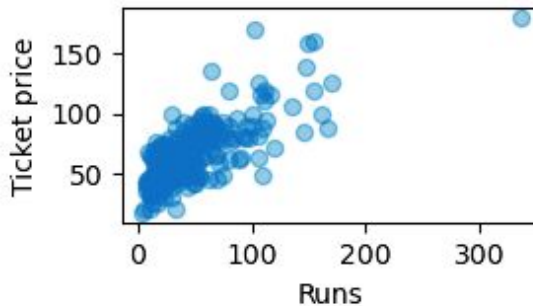
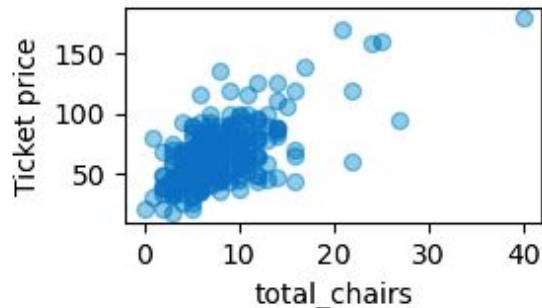
Recommendation/Key Findings

Recommendations:

- Ticket prices should be raised from the current \$81 to \$83
- The newly built lift and run extend the vertical drop by 150 ft
- Close down the least used run.

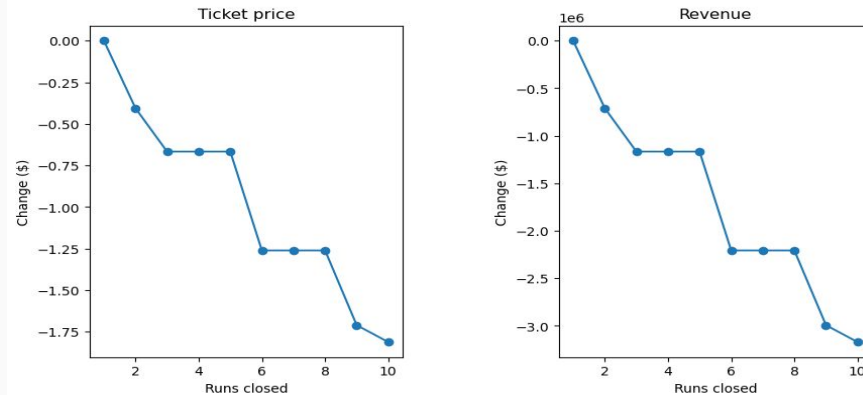
Modeling Results and Analysis

- Exploring the data we made scatter plots that plotted resort features and found that the features of total number of runs, chairs, vertical drop distances, and days open had fairly strong positive correlations with the ticket pricing. Thus features to focus on for the subsequent models.



Modeling Results and Analysis

- We have a predicted amount of visitors to be around 350,000. Shutting down the lifts and what it would do for pricing and revenue is shown in the graphs below. Shutting down one lift would not have much of an impact on the support for ticket prices and revenue. Any more than that would not be recommended.



Modeling Results and Analysis

- 2 more scenarios were ran that included adding a run, increasing the vertical drop by 150 ft, and installing an additional chair lift. 1 of those scenarios includes adding 2 acres of snow making. Both scenarios increase support for ticket prices of \$2 and the revenue to be over \$3.4M.
- One last scenario of just adding the longest run by 0.2 miles while adding snow coverage of 4 acres doesn't make any difference. Plus the linear and random forest regression models used have this feature pretty down on the feature importance list.

Summary and Conclusion

- As it is right now, our current ticket price of \$81 should be bumped up to \$83. This will be enough to support adding a run with a supporting lift that extends the resort's vertical drop by 150 ft. This will also give room to close down one of the previously least used runs.
- This would be a good start for us. However, obtaining more data such as visitor volume for other resorts and additional operating costs would help improve the model.

